WEEK 3 DELIVERABLE REPORT: D1 — PROJECT PROPOSAL

AgriUAV Platform: Precision Drone Dispatching Platform in the Mekong Delta

Project Metadata & Overview

Subject	ITA301 — Information System Analysis & Design
Deliverable	D1 — Project Proposal (15%)
Submission Date	End of Week 3 — 2026-06-02
Course Instructor	Mr. Tran Thanh Nguyen
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1. Problem Statement

Agricultural cooperatives and drone service enterprises in the Mekong Delta currently face a critical lack of digital tools to schedule, dispatch, and monitor spraying drone fleets, resulting in the following measurable consequences:

High Operational Costs: Managing spraying schedules manually via Zalo, Excel, or paper notebooks causes schedule conflicts, overlapping shifts, and wasted deadhead flights, inflating operational costs by 15–20%.

Low Field-Level Efficiency: Without accurate field mapping, drones spray uniformly across the entire field rather than concentrating variable dosages on specific pest-infected zones.

Safety Risks & Heavy Losses: The absence of automated safety geofencing has led to incidents where drones drifted off-course, causing neighbor crop poisoning, or crashed into 110kV high-voltage power lines (e.g., the Long An incident causing blackouts for 76,000 customers).

Compliance Hardships: No automated flight logging tools exist, failing to satisfy mandatory reporting requirements regarding drone identification numbers and pilot certifications specified under Decree 288/2025/ND-CP (effective November 5, 2025).

1.1. Proposed Solution

The AgriUAV Platform is designed as a Web and Mobile application ecosystem to automate scheduling, optimize drone dispatching via digital field mapping, calculate chemical dosages automatically, and track spraying progress in real-time.

1.2. System Scope

Operation management of a drone fleet scaled up to a maximum of 50 drones per cooperative within the rice cultivation regions of the Mekong Delta. The system excludes direct hardware flight control interception and embedded computer vision AI for pest detection (these are handled by independent peripheral systems).

1.3. In-Scope — System Functions

- Drone fleet management: drone identification codes, model specifications, tank capacity, and operational status (Available / In-Flight / Under Maintenance)

- Pilot profile management: certifications, licenses, and flight history records compliant with Decree 288/2025/ND-CP
- Spray scheduling: farmers and HTX managers submit spray requests; system confirms and records assignments
- Automated drone dispatch: assignment logic based on battery level, current GPS position, and schedule availability
- Real-time spray progress tracking on digital field map: drone position overlay and percentage area completion
- Automated alerts: low pesticide level, low battery, adverse weather (wind speed > 3 m/s), and geofencing boundary breach
- Pesticide inventory management and spray formula records per approved chemical registry
- Spray log export and seasonal compliance reporting for the Plant Protection Department
- Role-based access control: Admin / HTX Manager / Operator / Farmer
- System scale: maximum 50 drones per cooperative, within rice cultivation zones in the Mekong Delta

1.4. Out-of-Scope — Explicitly Excluded

- Direct drone hardware control from software (fly-by-wire or autopilot command interfaces)
- AI-based pest detection from drone imagery (computer vision or deep learning models)
- Integration with irrigation systems or other agricultural machinery
- Financial management, accounting, or VAT invoicing modules
- Online flight permit application with the Operations Department (UTM system integration)
- Physical cloud deployment or real-hardware drone integration (outside ITA301 scope — design only)

2. Business Case & ROI Analysis

2.1. Ecosystem Analysis — 3-Layer Model

To demonstrate the market gap that AgriUAV Platform will fill, the domain structure is categorized into three explicit layers:

Ecosystem Layer	Current Market State	AgriUAV Platform Intervention
Hardware Layer	Highly developed and saturated — DJI Agras (~80% Vietnam market share), XAG (~15%), YANMAR, and domestic brands (~5%), equipped with RTK positioning and multispectral cameras. Hardware availability is not a bottleneck.	Agnostic integration via mobile GPS telemetry at MVP stage. Planned native MAVLink telemetry protocol integration in future versions.
Service Layer	Local cooperatives (HTX) and private entities provide bulk drone spraying services (~150,000 VND/ha). Internal operations remain purely 'omni-manual' — managed via Zalo, Excel, and paper logbooks.	Provides the end-to-end digital coordination layer to automate scheduling, dispatch, and compliance for existing service providers.
Platform Layer	CURRENTLY VACANT — Total absence of a localized IS to optimize fleet dispatching, manage pilots, schedule trips, and maintain regulatory compliance logs. Pure entry point for an IS solution.	AgriUAV Platform fills this strategic gap, serving as the enterprise-grade operational coordination system for drone service cooperatives.

2.2. Market Sizing (TAM / SAM / SOM)

Based on verified field statistics as of June 2025 and empirical research, market sizing is grounded in actual drone counts rather than projections:

Segment	Drone Count & Data Source	Target Customer Profile	Strategic Rationale
TAM (Total Addressable Market)	~4,000 agricultural drones across 1.5 million ha of rice cultivation in the Mekong Delta.	Full Mekong Delta drone fleet	No digital platform in market = entire fleet is addressable
SAM (Serviceable Addressable Market)	1,215 active drones in Kien Giang (690), An Giang (265), Dong Thap (260). Source: Thanh Nien, June 2025.	HTX and commercial drone service operators in 3 core provinces	Operationally reachable in Year 1 rollout
SOM (Serviceable Obtainable Market)	200 drones managed by pilot-partner cooperatives linked to Can Tho University research network.	Strategic pilot HTX partners	Direct partnership channel for initial deployment

Revenue Model: B2B SaaS licensing at 500,000 VND / drone / month. Projected annual recurring revenue from SOM: 200 × 500,000 × 12 = 1,200,000,000 VND/year.

2.3. Cost-Benefit Analysis (Reference: 1 HTX — 10 drones, 500 ha/season)

Cost Category	AS-IS (Manual Operations)	TO-BE (With AgriUAV Platform)	Estimated Savings
Pesticide wastage (imprecise spraying)	~45 million VND/season	~31.5 million VND/season	–30% = 13.5M VND
Drone fuel (duplicate flights, wrong zones)	~12 million VND/season	~9.6 million VND/season	–20% = 2.4M VND
Coordination & reporting labor	~8 million VND/season	~2 million VND/season	–75% = 6.0M VND
Legal penalty risk (Decree 288 non-compliance)	Unquantified — ongoing exposure	Significantly reduced	Compliance built-in
TOTAL ESTIMATED SAVINGS / SEASON			**~21.9 million VND**

Platform operating cost: ~1,200,000 VND/month (cloud + API). For a 3-month season: 3,600,000 VND cost vs. 21,900,000 VND savings → Positive ROI from the first season of deployment.

3. Stakeholder Analysis

The system serves stakeholders across three tiers. Two actors marked (*) were supplemented by the team after identifying gaps in the initial AI-generated stakeholder list.

Classification	Stakeholder Group	Core Role & Expectations	Power (1–5)	Interest (1–5)	Strategic System Action
Primary	**Cooperative Management (HTX)**	Optimize fleet dispatching, lower operational costs by 15–20%, eliminate unverified pilot activity and pesticide leakage.	5	5	Intuitive Web Dashboard for scheduling, shift allocation, and financial tracking.
Primary	**Operators / Drone Pilots**	Receive clear flight orders, view digital field boundaries, navigate flight paths safely away from power lines, and log verified flight hours.	2	5	Lightweight Mobile App with GPS navigation and automated field check-in/check-out logging.
Primary	**Rice Farmers**	Easily book a spray service, track real-time drone completion status, and verify actual pesticide usage. Avg. age ~51 (Nguyen 2026), limited IT experience.	3	4	Automated SMS/Zalo notifications updating order status in real-time.
Primary	**Agricultural Engineers / Agronomists**	Define and validate spray formulas per crop type, manage approved pesticide registry, and confirm correct dosage application.	3	4	Formula management module with pesticide registry integration per Plant Protection Dept approved chemical list.
Secondary	**Agricultural Exporters (Trung An, Minh Phu)**	Trace pesticide application history data to meet strict clean grain export standards (EU, US) — VietGAP/GlobalGAP.	3	4	Exportable flight log modules matching VietGAP/GlobalGAP data standards.
Secondary	**Drone Manufacturers (DJI, XAG)**	Expand market reach through platform compatibility; benefit from telemetry integration via MAVLink SDK.	2	3	MAVLink telemetry API integration roadmap (planned post-MVP).
Secondary	**Banks / Agricultural Credit Institutions**	Use verified seasonal production data as a basis for agricultural lending risk assessment.	2	2	Seasonal operation summary export with verified flight logs usable as credit evidence.
Secondary (*)	**Pesticide Retailers**	Supply approved chemical inputs to HTX; inventory sync prevents unapproved pesticide use. [Supplemented by team]	2	2	Chemical inventory sync API connected to approved chemical registry.
Regulatory	**Plant Protection Dept. (Cục Trồng trọt & BVTV)**	Monitor compliance within safe airspaces and check approved pesticide databases under Decree 288/2025.	5	3	Automated report generation (PDF/Excel) linking drone serial numbers and pilot licenses.
Regulatory (*)	**Operations Dept. / General Staff (Cục Tác chiến)**	Primary military authority for UAV flight permits and drone ID registration under Decree 288/2025/ND-CP. [Supplemented by team]	4	2	Store and display device tracking UUIDs and certified pilot license numbers in compliance reports.
Regulatory	**Provincial Authorities**	Grant operational licenses for drone service providers	3	3	Periodic provincial spraying area reports

Classification	Stakeholder Group	Core Role & Expectations	Power (1–5)	Interest (1–5)	Strategic System Action
	(UBND / Sở NN&PTNT ĐBSCL)	at provincial level; receive periodic area-sprayed reports.			exportable to Ministry of Agriculture.

(*) Supplemented by team — not identified in initial AI stakeholder output. Verified against Decree 288/2025/ND-CP (Operations Dept.) and cooperative supply chain domain knowledge (Pesticide Retailers).

4. AS-IS Process Analysis

The current manual operational workflow is broken down into 5 phases with their corresponding pain points:

Seq	Operational Phase	As-Is Manual Workflow	Measurable Pain Points
1	**Demand Capture**	Farmers call or text the cooperative via Zalo to declare field areas needing spraying.	Ambiguous Boundaries: Lack of GPS coordinates leads to spraying the wrong fields, causing accidental neighbor crop exposure and boundary disputes.
2	**Scheduling & Assignment**	Management writes details in a paper journal or logs them in a shared Excel file, then calls pilots individually.	Double-Booking & Deadhead Flights: Scheduling conflicts cause overlapping shifts and misallocated dispatch routes, inflating operational costs by 15–20%.
3	**Field Execution**	Pilots bring the drone to the field and spray based on intuition or standalone hardware controls with no centralized telemetry.	Unverified Operator Activity: Operators may under-dose chemical applications or fabricate flight records to claim fuel and chemical allocations without detection.
4	**Monitoring & Control**	No active tracking infrastructure exists. Management waits at the office until the pilot verbally confirms completion by phone.	CRITICAL SAFETY RISK: Drones collide with high-voltage power lines due to lack of hazard visibility (Long An 110kV incident — 76,000 customers affected), or drift off-course, poisoning nearby plots.
5	**Reconciliation & Reporting**	At month-end, the cooperative tallies paper logs to invoice farmers and generates handwritten government reports.	Compliance Failure: Error-prone manual data cannot comply with Decree 288/2025/ND-CP, which mandates digital logs of drone IDs and pilot credentials. No verifiable audit trail exists.

5. Work Breakdown Structure (WBS)

WBS	Deliverable / Task	Member	Week	Specific Output
1.0	**D1 — Project Proposal**	**All**	**W1–W3**	**Complete proposal file**
1.1	**Choose topic, analyze the UAV domain**	**BA + PM**	**W1**	**Preliminary list of groups and topics**
1.2	**Stakeholder mapping + AS-IS analysis**	**BA**	**W2**	**Stakeholder map + Business case draft**

WBS	Deliverable / Task	Member	Week	Specific Output
1.3	Business case, cost-benefit, ROI	BA + PM	W3	Business case with specific data
1.4	WBS + Timeline + Risk Register	PM	W3	WBS table, preliminary Gantt chart, risk log
1.5	Internal peer review + completion of D1	All	W3	☑ Submit D1
2.0	D2 — SRS Document	BA + SA	W4	SRS following IEEE 830
2.1	Role-play interview stakeholders	BA	W4	Interview notes
2.2	Elicit & classify ≥16 FR + ≥7 NFR	BA + SA	W4	FR/NFR table with priority ratings
2.3	Write ≥10 Use Cases + User Stories	SA	W4	UC description + acceptance criteria
2.4	Complete SRS according to IEEE 830	BA + SA	W4	☑ Submit D2
3.0	D3 — System Modeling Package	SA	W5–W6	DFD + UML + ERD
3.1	DFD Level 0 + Level 1	SA	W5	Balanced DFD diagrams
3.2	Class Diagram + ≥3 Sequence Diagrams	SA	W5–W6	UML diagrams in StarUML
3.3	ERD 3NF + Data Dictionary	SA + SD	W6	Logical + Physical ERD
3.4	Traceability matrix SRS ↔ Models	SA	W6	☑ Submit D3
4.0	D4 — System Design Document	SD	W7–W9	Complete SDD
4.1	UI Wireframes ≥5 screens (Figma)	SD + QA	W7	Mobile app + web dashboard
4.2	Architecture + 2 alternatives + trade-off	SD	W8	Layered vs Microservices analysis
4.3	Physical data schema + security model	SD	W9	PostGIS schema + threat model
4.4	QA plan + test cases + ops plan	QA	W9	☑ Submit D4
5.0	D5 — Final Presentation & Defense	QA + All	W10	Slide + Demo + Q&A

6. Project Timeline

Week	Phase	Main Activities	Deliverable	Milestone
W1	P1 — Business	Kickoff, domain selection, team formation, stakeholder screening.	Topic proposal	Groups + domains confirmed
W2	P1 — Business	Stakeholder map, AS-IS analysis, pain points.	Business case draft	Complete Stakeholder Map
W3	P1 — Business	Business case, WBS, risk register, peer review.	☑ D1 Proposal	SUBMIT D1 — end of W3

Week	Phase	Main Activities	Deliverable	Milestone
W4	P2 — Requirements	Interview, elicit FR/NFR, write UC + SRS.	☑ D2 SRS	SUBMIT D2 — end of W4
W5	P3 — Modeling	PT1 (S25), DFD L0+L1, Class Diagram draft.	PT1 + DFD draft	Progress Test 1
W6	P3 — Modeling	Complete UML, ERD, and traceability matrix.	☑ D3 Models	SUBMIT D3 — end of W6
W7	P4 — Design	UI wireframes Figma, persona, user journey.	UI wireframes	≥5 complete screens
W8	P4 — Design	PT2 (S43), Architecture pattern, tech stack.	PT2 + Architecture	Progress Test 2
W9	P4 — Design	Trade-off analysis, security model, QA plan, ops.	☑ D4 SDD	SUBMIT D4 — end of W9
W10	P5 — Defense	PT3 (S55), mock defense, final presentation.	☑ D5 Defense	FINAL PRESENTATION — W10

7. Risk Register

#	Risk	Probability	Impact	Level	Mitigation Measures
R1	Scope creep — extending too many modules	High	High	Critical	Keep the out-of-scope frozen; each new feature requires PM approval before inclusion.
R2	Lack of factual data from cooperatives/farmers	Medium	High	High	Use data from Cục Trồng trọt reports, Nguyen et al. (2026), and verified press reporting as substitutes.
R3	Member takes unexpected leave mid-term	Low	High	High	Fully documented tasks on Notion; each deliverable has a designated backup member.
R4	Inconsistency between SRS ↔ UML ↔ SDD	High	High	Critical	Traceability matrix updated and reviewed after each deliverable submission.
R5	Lack of time to complete D4 (Week 9)	Medium	Medium	Medium	Begin architecture design from W7; do not allow work to accumulate into W9.
R6	Legal basis of Decree 288/2025 is unclear, making compliance module design difficult	Medium	Low	Medium	Scope compliance module at PDF report output level — not directly integrated with the Operations Department.
R7	Regulation change risk — NĐ 288/2025 may be supplemented by additional decrees	Low	Low	Low	Record assumption: 'Design is based on NĐ 288/2025 as of analysis date — requires review if new decrees are issued.'

#	Risk	Probability	Impact	Level	Mitigation Measures
R8	Domain knowledge gap — team lacks direct field experience in ĐBSCL	Low	Medium	**Medium**	Compensate via peer-reviewed literature (Nguyen 2026), field reporting, and role-play stakeholder interviews in W4.

8. Team Roles & Responsibilities

Member	Role	Main Deliverable	Notes
Dang Thanh Cong	Project Manager — planning, progress tracking, team coordination	Summary of all deliverables	Contact the instructor; maintain traceability matrix
Nguyen Thi Thanh Thao	Business Analyst — business analysis, stakeholder mapping, requirements elicitation	D1 Proposal, D2 SRS	Lead stakeholder interviews and FR/NFR elicitation
Le Van Minh	System Analyst — DFD, UML, ERD, and traceability models	D3 System Models	Use StarUML for all diagrams
Tran Anh Khoa	System Designer — architecture, UI/UX Figma, data design, security	D4 SDD	Use Figma for wireframes; PostGIS for data schema
Dang Vo Thanh Hieu	Presenter / QA — slides, storytelling, quality control, Q&A preparation	D5 Presentation	Cross-check consistency across all deliverables

9. D1 Quality Checklist

According to the Phase 1 checklist for the ITA301 course:

Status	Criteria	Completion	Reference
☑	Stakeholders mapped (Primary / Secondary / Regulatory) with Power & Interest ratings	Complete	Section 3
☑	Measurable problem statement with quantified impact (15–20% cost, 76,000 affected customers)	Complete	Section 1
☑	Clear scope — In-scope (10 functions) and Out-of-scope (6 exclusions) explicitly listed	Complete	Section 1.3–1.4
☑	Business case with specific data — 3-layer ecosystem, TAM/SAM/SOM, cost-benefit table, ROI	Complete	Section 2
☑	WBS + Timeline + Risk Register (≥5 risks with likelihood, impact, mitigation)	Complete	Sections 5, 6, 7
☑	AI Audit Log entries #01–#08 included (W1–W3 phase coverage)	Complete	Section 10

10. AI Artifact Audit Log — Entries #01–#08 (W1–W3 / Phase 1)

This consolidated audit log covers all AI interactions for Phase 1 (Problem & Business). Entries continue from W1 (#01–#03) and W2 (#04–#06) into W3 (#07–#08). The cumulative log will continue through D5, targeting ≥25 entries by end of W10.

No. (Week)	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
01 W1 / P1	**Problem & Business Decomposition**	Claude	"We are developing an IS project for ITA301 on Agricultural UAVs in the Mekong Delta. Suggest 3 distinct sub-domain directions for analysis, clearly stating the primary business problem and implementation complexity for each."	3 directions: (1) Fleet Management — generic fleet tracking; (2) Precision Spraying Platform — localized scheduling; (3) Marketplace — hourly drone rentals. All rated ★★★ complexity.	☑ SELECTED (2) Precision Spraying Platform: • Empirical data exists for Mekong Delta (Nguyen et al. 2026, N=940). • Pain points validated: HTX manages via Zalo/Excel. • Platform layer is entirely vacant in ecosystem. ✖ REJECTED (3): Scope too broad, no IS depth. ✖ REJECTED (1): Overlaps with another team's confirmed topic.	Ecosystem gap confirmed: Lao Dong (2025) — 'digital scheduling problem remains unresolved in ĐBSCL cooperatives.'
02 W1 / P1	**Pattern Recognition**	ChatGPT	"Compare Agricultural UAV, Delivery UAV, and Surveillance UAV across: (a) stakeholder complexity, (b) regulatory burden in Vietnam, and (c) data availability for an IS academic project."	Comparison matrix concluded Delivery UAV is 'easiest' due to lower perceived regulation. Agricultural UAV ranked as medium complexity.	⚠ AI RANKING REJECTED — incorrect evaluation: Agricultural UAV has a more structured legal framework (Decree 288/2025/ND-CP + Plant Protection Dept) than Delivery UAV in Vietnam. ☑ Maintained Agricultural UAV: (a) peer-reviewed research available (Nguyen 2026), (b) verifiable field incidents, (c) richer stakeholder map for IS analysis.	Decree 288/2025/ND-CP cross-checked via thuvienphapluat.vn — confirms specific UAV ID and pilot certification logging requirements applicable to agricultural UAV operators.
03 W1 / P1	**Decomposition — Ecosystem Mapping**	Claude	"For the AgriUAV Platform (ITA301, Mekong Delta): decompose the current Agricultural UAV ecosystem into structural layers. Identify which layers are occupied and which	AI proposed a 2-layer split: Hardware/Device layer and Application/User layer. Suggested the	✖ REJECTED 2-layer model — oversimplified and inaccurate. ☑ Team defined 3-Layer Model:	3-layer model validated: Thanh Nien (June 2025) confirms 1,215 active drones in 3 provinces with no platform-layer

No. (Week)	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
			represent market gaps for an IS solution."	gap lies in 'application integration.' No distinct Service Layer or Platform Layer identified.	• Hardware Layer (DJI/XAG — occupied) • Service Layer (HTX fleets — occupied, manual) • Platform Layer (digital IS coordination — VACANT) Reflects actual business reality confirmed by field reporting.	management software reported.
04 W2 / P1	**Decomposition — Stakeholder Mapping**	Claude	"Project: AgriUAV Platform for drone spraying in Mekong Delta. List all stakeholders, classify as Primary/Secondary/Regulatory, and outline core roles and expectations for each."	Suggested 6 general groups: Farmers, HTX, Pilots, Drone Manufacturers (DJI), Software Developers, and Local Government — totaling ~12 stakeholders.	☑ Accepted 5 core stakeholder groups. ✕ Removed 'Software Developer Team' — this is the project team, not an external stakeholder. + SUPPLEMENTED 2 critical actors AI omitted: • Operations Dept. / General Staff — sole authority for UAV flight permits under Decree 288/2025. • Pesticide Retailers — essential for inventory/compliance integration. + Split generic 'Government' into 2 specific regulatory bodies: Plant Protection Dept. + Operations Dept.	Operations Dept. authority confirmed: Vietnam Naval News — 'all flight permit records must be submitted to Cuc Tac chien, causing administrative delays.'
05 W2 / P1	**Decomposition — AS-IS Process Analysis**	Claude	"Decompose the AS-IS business process of an agricultural cooperative in Vietnam executing drone spraying services. List each step, the actor responsible, and pain points at each stage. Context: currently managed via Zalo/Excel."	Proposed a basic 4-step generic workflow: Intake → Dispatch → Execution → Reporting. Identified high-level pain points: lack of tracking, slow report compilation.	☑ Accepted 4-step structure as baseline. + Expanded to 5 steps — added 'Monitoring & Control' as a distinct phase (AI merged this into Execution). + Localized pain points with field-verified data:	110kV incident: Dan Viet (2024) — drone collision in Long An Province, 76,000 customer blackout. Decree 288 text confirms drone ID and pilot credential logging requirements.

No. (Week)	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
					• Phase 3: Operators may under-dose chemicals or inflate flight hours — AI missed this risk entirely. • Phase 4: 110kV power line collision in Long An (Dan Viet 2024) as concrete geofencing evidence. • Phase 5: Linked compliance failure directly to Decree 288/2025/ND-CP.	
06 W2 / P1	**Process / Algorithm — Business Case**	Claude	"Provide an industry-standard business case structure for an IS research project, including a preliminary ROI template and a formal problem statement formulation suitable for a university-level IS analysis project."	Standard 5-part structure (Context → Problem → Impact → Solution → Scope) and a generic ROI formula: (cost saved / investment) × 100%.	☑ Adopted structural blueprint as foundation. ☑ Applied Mekong Delta data localization: • 15–20% operational cost figure from Nguyen et al. (2026), N=940. • 1,215 drones across 3 provinces from Thanh Nien June 2025. • Added HTX Cost-Benefit table: pesticide –30%, fuel –20%, labor –75% = ~21.9M VND savings vs. 3.6M VND platform cost → positive ROI from Season 1. • Established 45% unmanaged area ratio as the addressable market segment.	15–20% cost figure: Nguyen et al. (2026). Province drone counts: Thanh Nien (June 2025). Cost-benefit: team derivation from published operational benchmarks.
07 W3 / P1 D1 Proposal	**Pattern Recognition — Business Case Comparison**	Claude	"Compare the business case of an Agricultural UAV platform vs. a Delivery UAV platform in Vietnam: market size, competition level, scalability, and suitability for IS research."	AI concluded Delivery UAV has 'a larger market' because logistics is booming. Agricultural UAV was rated lower on 'tech sophistication.'	⚠ COUNTER-ARGUMENT AGAINST AI BIAS: AI conflated 'market size = revenue potential' with 'research quality.' ITA301 evaluates domain richness, not revenue. Agricultural UAV is more suitable because:	Pattern recognition: team identified AI bias toward 'market = revenue' framing vs. ITA301 standard of 'domain richness = research quality.' No correction needed to deliverable.

No. (Week)	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
					1. Peer-reviewed paper exists (Nguyen et al. 2026, RWAE Journal) — no equivalent for Delivery UAV in Vietnam. 2. Real field data available by province from Vietnamese press. 3. Richer, more diverse stakeholder map (farmers + HTX + Cục BVTV + Cục Tác chiến). 4. 'Platform layer vacant' claim provable via 3-layer ecosystem decomposition. ☑ Team maintained Agricultural UAV selection. AI bias pattern recognized and documented.	
08 W3 / P1 D1 Proposal	**Decomposition — WBS & Risk Register**	Claude	"Propose a Risk Register for a 10-week IT project analyzing agricultural drone spraying in ĐBSCL. List 8–10 risks with likelihood (H/M/L), impact, and mitigation strategy."	AI listed 10 risks: scope creep, timeline, team conflict, data quality, tool learning curve, stakeholder availability, budget, integration complexity, security breach, deployment failure.	☑ Accepted 5 generic risks (scope creep, timeline, team conflict, stakeholder availability, data quality). ✖ REMOVED 'deployment failure' — ITA301 does not require actual deployment. ✖ REMOVED 'budget' — not applicable to a university project. ➕ SUPPLEMENTED 4 UAV-domain-specific risks AI missed: • Regulation change risk: NĐ 288/2025 newly effective, supplementary decrees possible. • Domain knowledge gap: team lacks direct field experience in ĐBSCL.	NĐ 288/2025 confirmed effective via VnBusiness and Government E-Portal. Risk R7 (regulation change) and R8 (domain gap) documented in Risk Register Section 7.

No. (Week)	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
					• Literature availability: peer-reviewed Agri UAV papers for Vietnam are limited. • Stakeholder interview difficulty: pilots/HTX managers are hard to contact in short timeframe.	

11. Verified Academic & Regulatory References

[1] Nguyen, D.L. et al. (2026). "The Effect of Farm Size on the Decision to Adopt Digital Technology: The Case of Unmanned Aerial Vehicles in Rice Production in Vietnam." Research on World Agricultural Economy. https://journals.nasspublishing.com/index.php/rwae/article/view/2358

[2] Government of Vietnam. Decree No. 288/2025/ND-CP — Regulatory Framework for Unmanned Aerial Vehicle Flights and Light-UAV Digital Registry (Effective: November 05, 2025). https://thuvienphapluat.vn/van-ban/Giao-thong-Van-tai/Nghi-dinh-288-2025-ND-CP-quan-ly-tau-bay-khong-nguoi-lai-va-phuong-tien-bay-khac-679996.aspx

[3] Vietnam News. (2026). Use of Agricultural Drones on the Rise in the Mekong Delta. https://vietnamnews.vn/society/1688198/use-of-agricultural-drones-on-the-rise-in-the-mekong-delta.html

[4] Lao Dong Newspaper. (2025). Digital Transformation Gaps in Mekong Delta Cooperatives: The Unresolved Software Problem. https://laodong.vn/xa-hoi/drone-nong-nghiep-va-bai-toan-so-hoa-o-dbscl-1498970.ldo

[5] Dan Viet Digital News. (2024). Investigation Report: Agricultural Drone Incident Involving 110kV Transmission Line Contact, Causing Outages for 76,000 Customers in Long An Province. https://tuoitre.vn/drone-phun-thuoc-tru-sau-va-vao-day-110kv-gay-mat-dien-o-5-huyen-tai-long-an-20241014173928997.htm

[6] Vietnam Naval News. (2024). Management of Unmanned Aerial Systems and Flight Permit Processes via the Operations Department (Cuc Tac chien). https://baohaiquanvietnam.vn/tin-tuc/quan-ly-thiet-bi-bay-khong-nguoi-lai-trong-nong-nghiep-khong-de-cong-nghe-tro-thanh-hiem-hoa

[7] Thanh Nien Newspaper. (June 2025). Province-Level Statistical Breakdown of Active Agricultural Drone Fleets in the Mekong Delta Region. https://thanhnien.vn/bung-no-uav-nong-nghiep-o-mien-tay-185250617232615926.htm

[8] CropLife Vietnam. (2025). Forum: Making Drones an Effective Assistant for Farmers. https://croplifevietnam.org/toa-dam-dua-drone-tro-thanh-tro-thu-dac-luc-cua-nha-nong.html